B0 Start here: foundations for data science

The Python, pandas and statistics this track assumes.

B0-L1 Python essentials for data work

Put logic in typed functions that return results, use comprehensions to filter, and test for missing values with `is None`.

```python
def mean_weight(weights: list[float]) -> float | None:
    """Average weight, or None for an empty list."""
    if not weights:
        return None
    return sum(weights) / len(weights)
```

Test for a missing number	if value is None. Truthiness also treats 0 and 0.0 as false.
Filter a list in one line	A comprehension: [w for w in weights if w > 0].
Show 0.125 as 12.5%	f"{share:.1%}".
Walk through a dict	for key, value in d.items():
Signal bad input	raise ValueError("message naming the input"). Catch it with try/except where you can handle it.

B0-L2 pandas essentials

Filter with bracketed masks, summarise with groupby and named aggregations, and merge with how="left" when every row must survive.

```python
import pandas as pd

df = pd.read_csv("respondents.csv")
```

```python
df.groupby("region", as_index=False).agg(
    n=("w", "size"),        # rows, missing or not
    recorded=("w", "count"),  # non-missing values
    total=("w", "sum"),
)
```

Filter rows on two conditions	df[(df["w"] > 0) & df["in_scope"]]. Each comparison in brackets, & not and.
Count rows against count values	size counts rows, count counts non-missing values. sum and mean skip missing values.
Several results per group	df.groupby("region", as_index=False).agg(n=("w", "size"), total=("w", "sum")).
Join and keep every row	df.merge(other, on="key", how="left"). The default inner merge drops rows with no match.

B0-L3 The statistics you need for this track

Report an estimate with its standard error or interval, know which SD your library computes, and read a p-value for what it is.

Standard error of a mean and of a share	Mean: SD / sqrt(n). Share: sqrt(p(1 - p) / n). Four times the sample halves the SE.
A rough 95% interval	Estimate plus or minus 1.96 standard errors.
What a p-value is	The probability of a difference at least this large if there were no true difference. Not the probability that a hypothesis is true.

np.std against pandas std	np.std divides by n (population). Series.std divides by n - 1 (sample). Use np.std(x, ddof=1) for the sample SD.